1. a) 7C02H = 0111 1100 0000 0010

TI = 0 (GDT)

DI = 1111 1000 0000 = F80H

b) End = Base + Limit * 2K

= 1F308810H + 78BD77FFH

= 97EE000FH

c) Access: 54H = 0101 0100

A = 0, so access is not granted.

d) DPL = 10

e) E = 0, ED = 1. So, SS.

f) Total number of rows in GDT & LDT

$= 2^{13} + 2^{13} = 2^{14} = 16,384$

Limit = 20bits. So each descriptor represents a memory segment of size(maximum) = $2^{20} * 2KB$

∴ Max virtual memory size = $2^{14} * 2^{20} * 2KB$

$= 2^{45}$

= 32TB

= 2^15 GB

2. (a) Duty cycle = $\frac{4}{5}$ = 0.8 = 80%

100% D.C. $\Rightarrow 2^7 - 1 = 127$

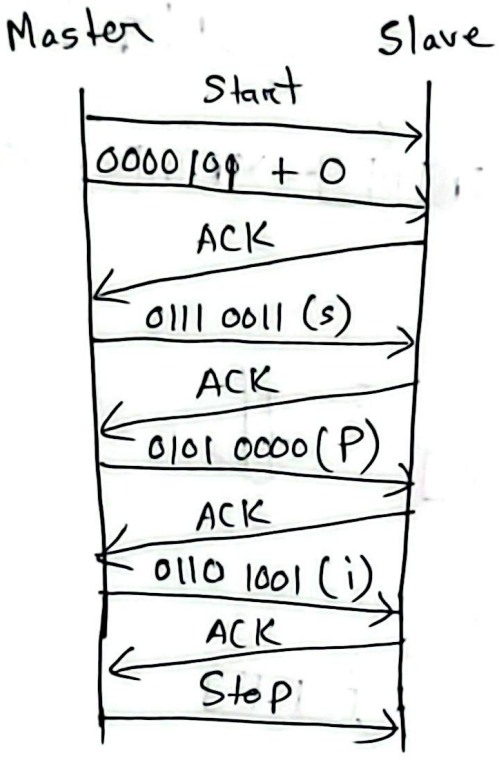
$\therefore$ 1% D.C. $\Rightarrow$ 127/100%

$\therefore$ 80% D.C. $\Rightarrow (127 \times 80\%)/(100\%)$

$= 101.6$

So, $x = 101.6$

(b) SPI is better for smaller communication system.



S.A. = 5H = 000 0101

s = 115 = 0111 0011

P = 80 = 0101 0000

i = 105 = 0110 1001

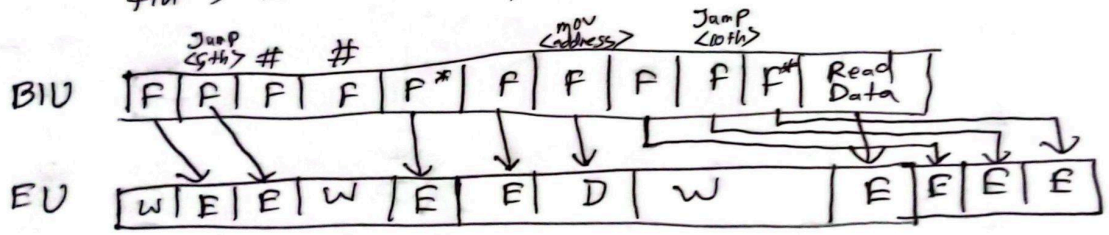
(c) Arduino would be more suitable for this task.

We are mainly using hardware devices (sensors and alert system) and doing the same task repeatedly. We do not need high processing power for this work. So, Arduino is more suitable here.

CamScanner

2nd → jump <5th> , 4th → memory read

7th → mov <address> , 9th → jump <10th>



Scanned with CamScanner

3(b) (i) $\frac{32\ KB}{4\ KB} = 8$ pages

(ii) 8 pages × 8 bytes

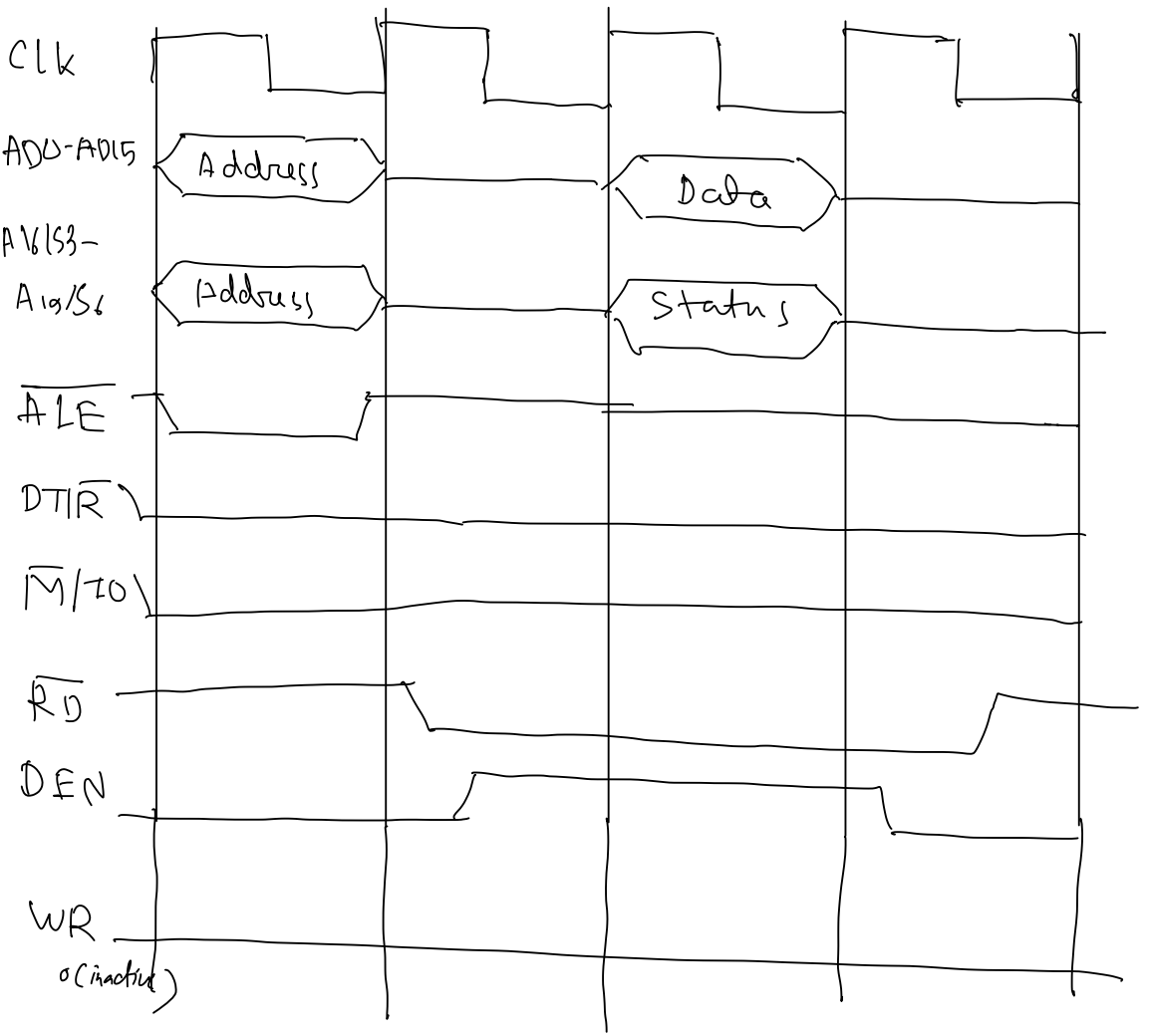
= 64 bytes

(3) Pages in primary memory = 5

∴ secondary storage = 8 − 5 = 3

∴ percentage = $\frac{3}{8} \times 100 = 37.5\%$

3(c) Read operation
Memory



4

a) expression: $-5 + 2X9A - 3^2$

$\equiv 5 - A + 9A + 9A - 3X3$

$= 5 - A + 9A + 9A - 3 - 3 - 3$

RAM content table:

Address	Assembly	Content
0H	SUB CH	0010 1100
1H	ADD 8H	0001 1000
2H	ADD 8H	0001 1000
3H	SUB EH	0010 1110
4H	SUB EH	0010 1110
5H	SUB EH	0010 1110
6H	OUT	1101 XXXX
7H	HLT	1011 XXXX

b)

Address	Assembly	Acc = 5H	B = 10H
0H	SUB CH	5H − AH = −5H	AH
1H	ADD 8H	−5H + 9A = 95H	9AH
2H	ADD 8H	95H + 9AH = 2FH	9AH
3H	SUB EH	2F − 3H = 2CH	3H
4H	SUB EH	2C − 3H = 29H	3H
5H	SUB EH	29H − 3H = 26H	3H
6H	OUT	26H	3H
7H	HLT	0H	0H

In 3rd instruction (at 2H), the sum is 12FH. But Acc Register holds only 8bits (from LSB).

c) $CON = L_I \overline{E_I} L_A \overline{E_A} \quad \overline{E_u} S_u L_o L_B \quad \overline{E_p} C_p \overline{L_M} \overline{CE}$

Disable $\Rightarrow$ 0 1 0 1 1 1 0 0 1 0 1 1

$= 5CBH$

i) $CON = 7EA$

$= 0\ 1\ \underset{L_A}{\boxed{1}}\ 1 \quad 1\ \underset{L_o}{\boxed{1}}\ 0 \quad 1\ 0\ 1\ \underset{\overline{CE}}{\boxed{0}}$

Operation: RAM sends the content (stored at 9H (MAR)) to Acc and Output Register.

ii) $CON = 4DB$

$= 0\ 1\ 0\ \underset{\overline{E_A}}{\boxed{0}} \quad 1\ 1\ 0\ \underset{L_B}{\boxed{1}} \quad 1\ 0\ 1\ 1$

Operation: Acc sends its content to B Register

iii) $CON = 72BH$

$= 0\ 1\ \underset{L_A}{\boxed{1}}\ 1 \quad \underset{\overline{E_u} S_u L_o}{\boxed{0\ 0\ 1}}\ 0 \quad 1\ 0\ 1\ 1$

Operation: Subtraction of Acc and B Register occurs, and result is sent to Acc and Output Regis

T-State	Acc = 5H	= 10H B Reg	Output Reg = 7H
T4	5H	10H	5H
T5	5H	5H	5H
T6	5H-5H=0H	5H	0H